Speaker-Independent Topic Classification of Spontaneous Dholuo Agricultural Speech Transcripts

Technical report

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Abstract

Dholuo is a Western Nilotic language spoken by millions of people in western Kenya and northern Tanzania, yet it has few labelled resources for natural language processing. This report studies whether transcribed, spontaneously spoken Dholuo answers about agriculture can be assigned to one of 23 agricultural sub-domains when every test speaker is unseen during training. Using the Utavu Foundation Dholuo Speech Dataset (1,359 approved transcripts from 23 speakers), we audit the data, build a speaker-disjoint split (810 / 188 / 250 transcripts from 13 / 5 / 5 speakers), and compare majority and TF-IDF baselines, a BiLSTM with attention, and the pretrained multilingual encoders AfroXLMR-large-114L and XLM-R-large under full fine-tuning, LoRA and a frozen encoder. Ten controlled experiments examine training-set size, normalisation, class weighting, speaker leakage, dialect, code-switching, fine-tuning strategy, prompt overlap and existing machine-translation tools. LoRA-tuned AfroXLMR obtains the highest mean test macro-F1 (0.436 over three seeds), but the five leading models are statistically indistinguishable, and a pre-registered “prefer the simpler model” rule selects word-level TF-IDF with logistic regression (test macro-F1 0.396, 95% interval 0.329 to 0.442). African-language pretraining is decisive at fixed architecture (AfroXLMR 0.416 against XLM-R 0.101). The largest effect is prompt overlap: speakers repeat a median 60% of the question’s words, and removing training prompts that also appear in the test set lowers macro-F1 from 0.396 to 0.311. A translate-to-English pipeline with NLLB-200 performs significantly worse (0.181 zero-shot, 0.318 trained on translations). The selected model is served in a Streamlit application that applies the exact training preprocessing and withholds confidence percentages until they are calibrated on validation data. We discuss the limits imposed by 23 speakers, question-derived labels and under-confident probabilities.

1. Introduction and background

Most NLP research and tooling concentrates on a small number of high-resource languages, leaving the majority of the world’s languages with little data or software support (Joshi et al., 2020). African languages are heavily affected, although community efforts have produced multilingual encoders adapted to African languages, such as AfriBERTa (Ogueji et al., 2021), AfroXLMR (Alabi et al., 2022) and SERENGETI (Adebara et al., 2023), and evaluation datasets such as MasakhaNEWS (Adelani et al., 2023) and SIB-200 (Adelani et al., 2024).

Dholuo (ISO 639-3 `luo`) belongs to the Western Nilotic branch and is spoken mainly around Lake Victoria. Its use is concentrated in informal settings, while English and Kiswahili dominate formal domains such as education, governance and commerce (Mbogho et al., 2025). For agricultural information services this creates a gap: farmers who are most comfortable speaking Dholuo are the least served by text-based systems in English or Kiswahili.

The Utavu Foundation Dholuo Speech Dataset, released in 2026 by the Utavu Foundation and CLEAR Global and stewarded by the Maseno Centre for Applied AI, contains spontaneous spoken answers to agricultural questions with editor-approved transcripts, speaker metadata and topic labels (Utavu Foundation and CLEAR Global, 2026). It makes it possible, for the first time to our knowledge, to study fine-grained agricultural topic classification of spontaneous Dholuo under speaker-independent conditions.

2. Problem definition

Research question. How accurately can classical, recurrent and pretrained multilingual models assign transcribed, spontaneously spoken Dholuo agricultural answers to their agricultural sub-domain when every test speaker is unseen during training, and how does that accuracy change with training-set size, text normalisation, dialect, code-switching, fine-tuning strategy, prompt overlap, and compared with existing machine-translation tools?

Element	Definition
Input	The editor-approved Dholuo transcript of one spoken answer (10 to 60 seconds of speech; 16 to 158 words)
Output	One of 23 agricultural sub-domains, with a ranking over all 23
Target	The `sub_domain` field of the free-form subset, restricted to classes with at least 25 recordings and 5 speakers
Primary metric	Macro-F1 on speaker-disjoint test data
Secondary metrics	Accuracy, macro precision and recall, weighted F1, per-class precision/recall/F1, confusion matrix, bootstrap 95% intervals, seed-to-seed variation, expected calibration error

An important property of the target must be stated at the outset: the label of each recording is the sub-domain of the **question** the speaker was asked, inherited from the prompt. It is not an independent annotation of the **answer**. The task is therefore to predict the question-associated sub-domain of a transcript; it does not establish the objectively correct topic of what was said. Sections 7.5 and 8 show where this distinction matters.

3. Objectives and scope

Objectives. (1) Audit the dataset and choose a defensible supervised task from the evidence. (2) Build reproducible preprocessing and a leakage-controlled, speaker-disjoint split. (3) Compare floors, classical baselines, a recurrent model and pretrained transformers under one protocol. (4) Run controlled experiments that each answer a stated

question. (5) Analyse errors quantitatively and qualitatively. (6) Deploy the selected model in a simple application that communicates its limits.

In scope. Text classification of transcripts from the free-form subset; Dholuo only; agriculture only; local deployment.

Out of scope. Speech recognition (the application expects text), translation, agricultural advice, detection of out-of-domain or non-Dholuo input, and public deployment or redistribution of data or models.

The choice of topic classification over automatic speech recognition was made from the audit (notebook section 4): every free-form transcript carries a label; the classical-recurrent-pretrained comparison applies directly, which it does not for ASR; and every experiment fits the Kaggle GPU budget with three seeds, whereas ASR fine-tuning would allow at most one seed per configuration. ASR on the same speaker split is proposed as future work.

4. Background and existing approaches

Pretrained encoders for African languages. XLM-R (Conneau et al., 2020) is a Transformer encoder pretrained on CommonCrawl text in 100 languages; Luo is not among them. AfroXLMR (Alabi et al., 2022) continues masked-language-model training of XLM-R on African languages (multilingual adaptive fine-tuning). The checkpoint used here, `Davlan/afro-xlmr-large-114L`, lists Luo among its pretraining languages on its model card. We make no assumption about how much Dholuo text it saw.

Topic classification for African languages. MasakhaNEWS (Adelani et al., 2023) provides news topic classification in 16 languages that do not include Luo. SIB-200 (Adelani et al., 2024) annotates about 1,000 Flores-200 sentences with 7 broad topics (science/technology, travel, politics, sports, health, entertainment, geography) and extends the labels to the 200+ Flores languages, which include Luo; its texts are written translations of English Wikipedia sentences rather than spontaneous speech, and none of its topics concerns agriculture.

Machine translation. NLLB-200 (NLLB Team, 2022) is an open translation model covering Luo (`luo_Latn`), and Google Translate added Dholuo in its 2024 expansion to 110 new languages, announced in June 2024 (Google, 2024). A natural alternative to a Dholuo classifier is therefore to translate into English and classify with English tools such as entailment-based zero-shot classification (Yin et al., 2019) using BART-large fine-tuned on MultiNLI (Lewis et al., 2020; Williams et al., 2018).

Parameter-efficient fine-tuning. LoRA (Hu et al., 2022) freezes a pretrained model and trains low-rank updates to selected weight matrices, reducing the number of trained parameters by orders of magnitude.

Gap. To our knowledge no public dataset or model addresses fine-grained agricultural topic classification of spontaneous Dholuo evaluated on unseen speakers. Experiment I (section 7.6) tests the strongest existing-tools route empirically rather than relying on this literature search alone.

5. Dataset description and data-quality analysis

5.1 Source and access

Property	Value
Dataset	Utavu Foundation Dholuo Speech Dataset, Hugging Face `MCAA1-MSU/twb_luo_ke`
Revision used	`5cbcd61d844e29e80e5e574fa14a7c54603052c3`
Creators	Utavu Foundation and CLEAR Global (TWB Voice platform); steward: Maseno Centre for Applied AI
Licence	Nwulite Obodo Open Data License (NOODL-1.0); gated access
Recording period	16 May to 16 June 2026 (dataset card)
Files used	`utavu-freeform_prompts_data/Files/transcripts.csv` and `meta.csv`; the image-prompt equivalents for statistics only

The repository layout (`Files/transcripts.csv`, `Files/meta.csv`) differs from the paths given in the dataset README (`files/freeform_transcripts.csv`, `files/freeform_metadata.csv`); the notebook locates the files from the repository listing.

5.2 Record-level quality

Check	Free-form subset	Image-prompt subset
Rows	1,359	1,029
Speakers	23	21 (all also in free-form)
Unique prompts	1,019 (340 answered by more than one speaker)	658
Audio	10.64 h; 10.1 to 59.8 s per clip	8.93 h
Missing values, empty transcripts, duplicate files or transcripts	none	none
Transcription status	all approved	1 rejected
Editor-corrected transcripts	586 (43%)	161 (16%)
Words per transcript	mean 59.9, range 16 to 158	mean 64.3, range 17 to 172
Speech-rate outliers (< 0.5 or > 4.5 words/s)	none (median 2.14 words/s)	none

5.3 Text-level quality

The text contains inconsistencies that matter for preprocessing: code-switching markers appear as `[cs]` (771 occurrences), `[CS]` and `[[cs]`; pause markers as `[pause]` and `[Pause]`; one transcript has an unclosed tag and one an odd number of `[cs]` markers; 144 transcripts lack a space after punctuation; 20 use curly apostrophes; and capitalisation is irregular. 225 free-form transcripts (17%) contain at least one code-switched span into English or Kiswahili.

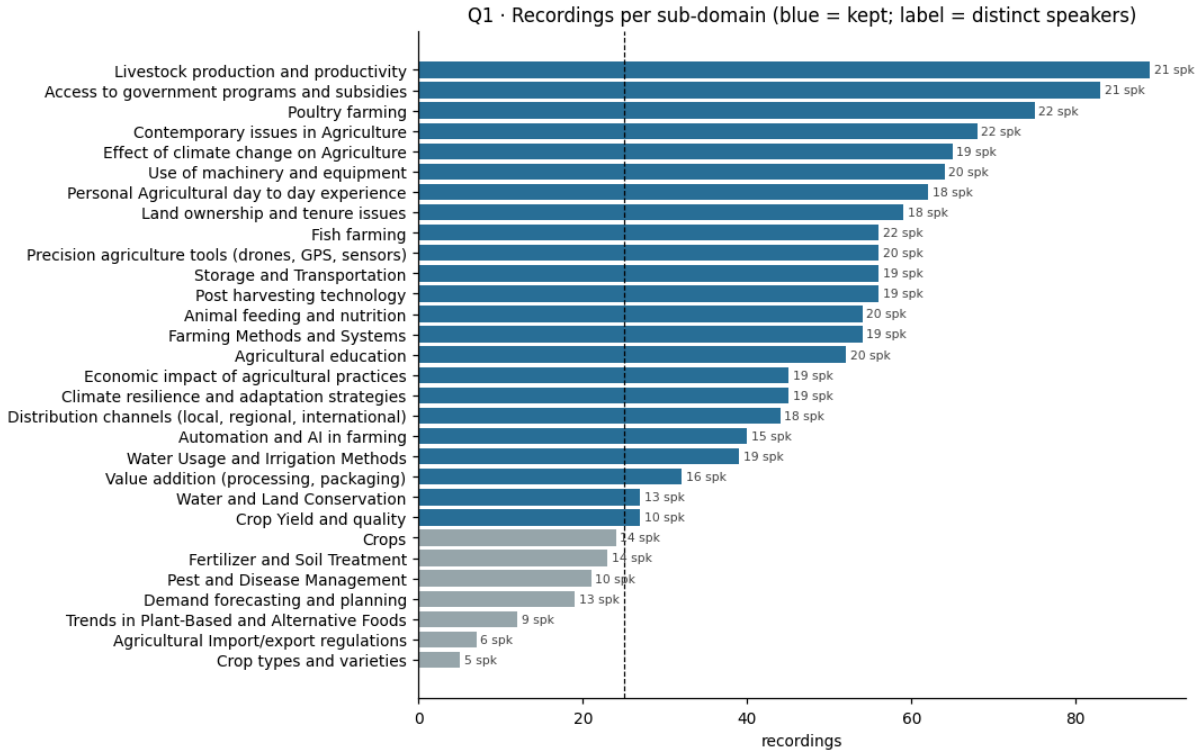
5.4 Speakers and dialect

Attribute	Distribution (23 speakers)
Gender	12 male, 11 female
Age	18 to 29: 18; 30 to 39: 4; 50 to 59: 1
Dialect (project survey)	Nyanduat (north): 14; Milambo (south): 9
Education	Bachelor’s 10; college diploma/certificate 9; secondary 3; postgraduate 1

The two dialect fields in the metadata disagree for 14 of the 23 speakers; for example, nine speakers who chose “Milambo” on the registration form report Nyanduat as their primary variety in the survey. Following the dataset card, we use the survey field, and we treat dialect results with caution.

5.5 Label validity

Of 30 sub-domains, 7 have fewer than 25 recordings (5 to 24) and are excluded, leaving **23 sub-domains and 1,248 transcripts** (111 excluded). Among kept classes the largest-to-smallest ratio is 3.3 (89 to 27 recordings), every class has at least 10 speakers, and no single speaker contributes more than 25% of a class (figure 3). Every prompt maps to exactly one sub-domain. Figure 1 shows the class sizes.

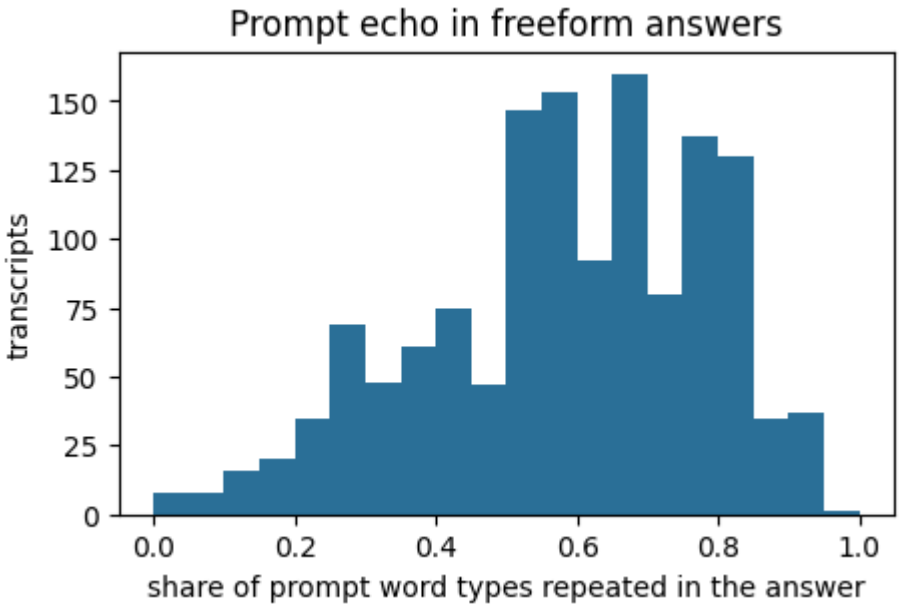


Recordings per sub-domain; blue classes are kept, labels give the number of distinct speakers.

5.6 Leakage risks and prompt echo

Three features could inflate scores: all recordings come from 23 people; the prompt text determines the label (it is never used as input); and prompts recur across speakers while answers repeat their prompts. We measured the last effect directly: **the median answer**

contains 60% of its prompt's word types, 33.5% of answers open by restating the prompt, and 14.9% contain at least 80% of the prompt's words (figure 2).



Share of each prompt's word types repeated in the answer.

6. Methodology and experimental design

6.1 Preprocessing and feature engineering

Cleaning steps (empty text, rejected transcription, duplicate file, duplicate speaker-transcript pair) removed no rows, consistent with the audit; only the 111 recordings in ineligible classes were dropped. Three text variants were built:

- `minimal`: Unicode NFC and whitespace normalisation;
- `normalized`: additionally straight apostrophes, lower case, consistent spacing around punctuation, and one canonical token each for code-switching (`<cs>`), pauses (`<pause>`) and unclear speech (`<unclear>`) whatever spelling the transcriber used;
- `normalized_notags`: as `normalized` with markers removed and code-switched words kept.

Stop-word removal, stemming, spelling correction, removal of apostrophes inside words (such as *ng'ado*) and removal of English or Kiswahili words were deliberately avoided because they discard linguistic information. The preprocessing was written once, executed in the notebook and exported with the model; the application uses an identical implementation whose parity with the notebook is tested automatically.

Features: TF-IDF over word 1-2-grams (word pattern keeps apostrophes and markers) or character 2-5-grams within word boundaries, with sublinear term frequency and case preserved (so that the variants genuinely differ); learned word embeddings for the BiLSTM; SentencePiece subwords for the transformers.

6.2 Speaker-disjoint split

With 23 speakers a random choice of test speakers can easily omit classes or a dialect. A seeded search over 20,000 candidate speaker groupings (seed 42) selected test speakers,

then validation speakers, minimising the Jensen-Shannon divergence between each split's class distribution and the overall distribution, with penalties for missing classes, deviation from 20% (test) and 15% (validation) of rows, and absence of either dialect. Only labels and metadata were used.

Split	Transcripts	Speakers	Nyanduat / Milambo speakers	Female speakers	Code-switched	Mean words
Train	810	13	10 / 3	6	113	52.0
Validation	188	5	2 / 3	2	38	79.6
Test	250	5	2 / 3	3	52	70.9

All 23 classes appear in every split. There is no speaker overlap and no exact or near-duplicate transcript across splits (character TF-IDF cosine above 0.9: none). By design, 101 of the 250 test answers respond to a prompt also answered by a training speaker; this is studied in Experiment H. Only 3 of the 13 training speakers speak Milambo.

6.3 Models

Model	Architecture and training
Majority, stratified	Constant most frequent class; random by class prior
TF-IDF + LR (word, char)	Multinomial logistic regression, L2, class-weighted; C and n-gram settings by grid search on validation
TF-IDF + SVM	Linear SVM on concatenated word and character TF-IDF; C by grid search; margins converted to probabilities with a temperature fitted on validation (T = 0.080)
BiLSTM + attention	128-d embeddings learned from scratch (training vocabulary, frequency at least 2), one bidirectional LSTM layer (2 x 128), additive attention pooling, dropout 0.4, linear output; AdamW (lr 2e-3), class-weighted cross-entropy, early stopping on validation macro-F1 (patience 6, up to 40 epochs); 563,095 parameters
AfroXLMR full fine-tuning	`Davlan/afro-xlmr-large-114L`, 24 layers, hidden size 1024, 16 heads, 250,002-piece vocabulary; all weights except the input-embedding matrix updated (303.9M of 559.9M parameters); AdamW lr 2e-5, linear warm-up (10%) and decay, effective batch 16, fp16, gradient checkpointing, up to 20 epochs with patience 5
AfroXLMR LoRA	Encoder frozen; rank-16 adapters (alpha 32, dropout 0.1) on attention query and value, plus classification head (2.65M trainable parameters); lr 2e-4
AfroXLMR frozen	Mean of final hidden states, logistic-regression head (23,575 parameters)
XLM-R full fine-tuning	`FacebookAI/xlm-roberta-large`, identical settings to AfroXLMR full fine-tuning

Freezing the input embeddings and enabling gradient checkpointing kept peak GPU memory at 7.2 GB on a 15.6 GB Tesla T4, and limits drift of rare-token embeddings on 810 examples. Both checkpoints use the same tokenizer and segment Dholuo identically (1.7 subwords per word; 44.8% of words are a single piece), so differences between them reflect pretraining rather than tokenization. Seeds 13, 42 and 2026 were used for every neural model. The complete hyperparameter table is in Appendix B.

6.4 Selection protocol and statistics

Test scores were recorded for every run but never used for any choice. Hyperparameters, the text variant and the final model were chosen on validation macro-F1. The final-model rule was fixed in advance: rank models by mean validation macro-F1; among models within one standard deviation of the best, choose the least complex (majority < logistic regression < SVM < BiLSTM < frozen encoder < LoRA < full fine-tuning). Confidence intervals use 2,000 bootstrap resamples of test rows and, more conservatively, of test speakers; model comparisons use a paired bootstrap on the same resamples; subgroup error rates use Wilson intervals (Wilson, 1927; Efron and Tibshirani, 1993).

6.5 Experiments

ID	Question	Design
A	How quickly do models improve with more data?	Per-class subsampling of training rows at 10, 25, 50, 75, 100%
B	Does normalisation help?	Three text variants for TF-IDF LR and BiLSTM
C	Does model sophistication pay off?	All models on one split, paired bootstrap
D	Does dialect matter?	Subgroups on test and on speaker-grouped out-of-fold predictions
E	Does code-switching cause failures?	Same subgroup design
F	How much does speaker leakage inflate scores?	5-fold speaker-grouped vs random-row CV on development speakers, three repetitions
G	Which fine-tuning strategy is best for the compute?	Frozen, LoRA, full fine-tuning on AfroXLMR
H	Is the model reading the answer or recognising the question?	Seen vs unseen prompts; retraining without test prompts
I	Do existing tools already solve the task?	NLLB-200 translation, then zero-shot BART-MNLI or TF-IDF LR
Ablation	Does class weighting help?	Weighted vs unweighted loss, C re-tuned for each

7. Results

All numbers below are taken from the executed notebook; Appendix A maps each table to its notebook section. The full run took 171.2 minutes on a Tesla T4 and logged 91 model runs.

7.1 Model comparison and selection (Experiment C)

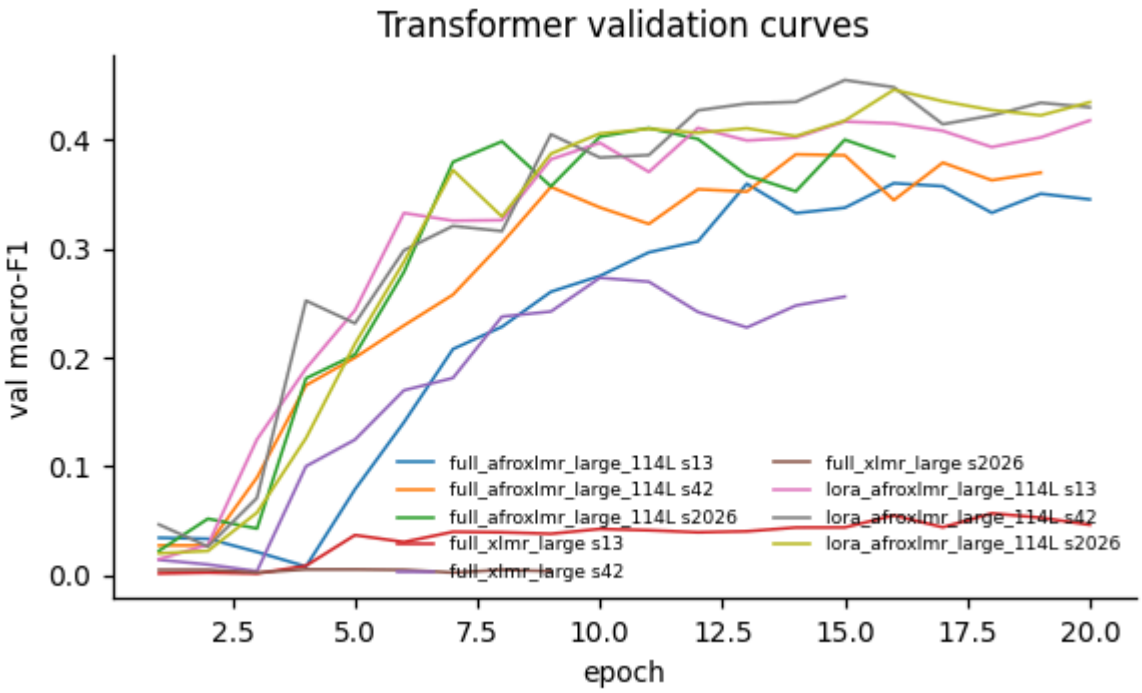
Model	Runs	Val macro-F1	Test macro-F1	Test acc.	Test weighted F1	95% CI, rows	95% CI, speakers
LoRA AfroXLMR	3	0.440 ± 0.019	0.436 ± 0.060	0.452	0.434	0.309 to 0.425	0.314 to 0.420
TF-IDF + SVM	1	0.432	0.413	0.448	0.414	0.345 to 0.459	0.367 to 0.425
TF-IDF + LR (word), selected	1	0.427	**0.396**	0.424	0.409	0.329 to 0.442	0.375 to 0.425
TF-IDF + LR (char)	1	0.423	0.371	0.408	0.373	0.306 to 0.414	0.329 to 0.396
AfroXLMR full	3	0.386 ± 0.025	0.416 ± 0.025	0.435	0.423	0.328 to 0.441	0.330 to 0.457
BiLSTM + attention	3	0.337 ± 0.027	0.307 ± 0.022	0.339	0.318	0.234 to 0.331	0.258 to 0.294
AfroXLMR frozen	1	0.290	0.227	0.256	0.239	0.177 to 0.266	0.178 to 0.267
XLM-R full	3	0.112 ± 0.142	0.101 ± 0.135	0.149	0.107	0.202 to 0.296	0.222 to 0.276
Stratified random	1	0.039	0.035	0.048	0.043	0.017 to 0.055	0.023 to 0.045
Majority class	1	0.005	0.005	0.056	0.006	0.002 to 0.007	0.003 to 0.006

Means and standard deviations are over seeds; the interval columns describe each model's best-validation run (the run that would be deployed), which is why the interval for LoRA and XLM-R does not centre on the mean.

LoRA AfroXLMR has the best validation score (0.440 ± 0.019). The SVM and both logistic regressions fall within one standard deviation, so the rule selects **word-level TF-IDF + logistic regression** (unigrams, C = 1). Paired bootstrap differences on the test set (selected model minus other):

Comparison	Difference	95% interval	Conclusion
vs LoRA AfroXLMR (best-val run)	+0.018	-0.044 to +0.082	tie
vs TF-IDF + SVM	-0.017	-0.061 to +0.031	tie
vs TF-IDF + LR (char)	+0.025	-0.026 to +0.072	tie
vs AfroXLMR full	-0.002	-0.067 to +0.067	tie
vs BiLSTM	+0.107	+0.039 to +0.172	selected model better
vs AfroXLMR frozen	+0.168	+0.107 to +0.224	selected model better
vs XLM-R full	+0.140	+0.074 to +0.202	selected model better

On 810 training examples the strongest pretrained model matches but does not beat a well-tuned linear model. African-language pretraining is decisive at fixed architecture and tokenizer: AfroXLMR full fine-tuning gave 0.407, 0.444 and 0.397 across seeds, while XLM-R gave 0.042, 0.256 and 0.005, the last run collapsing to near-constant predictions.



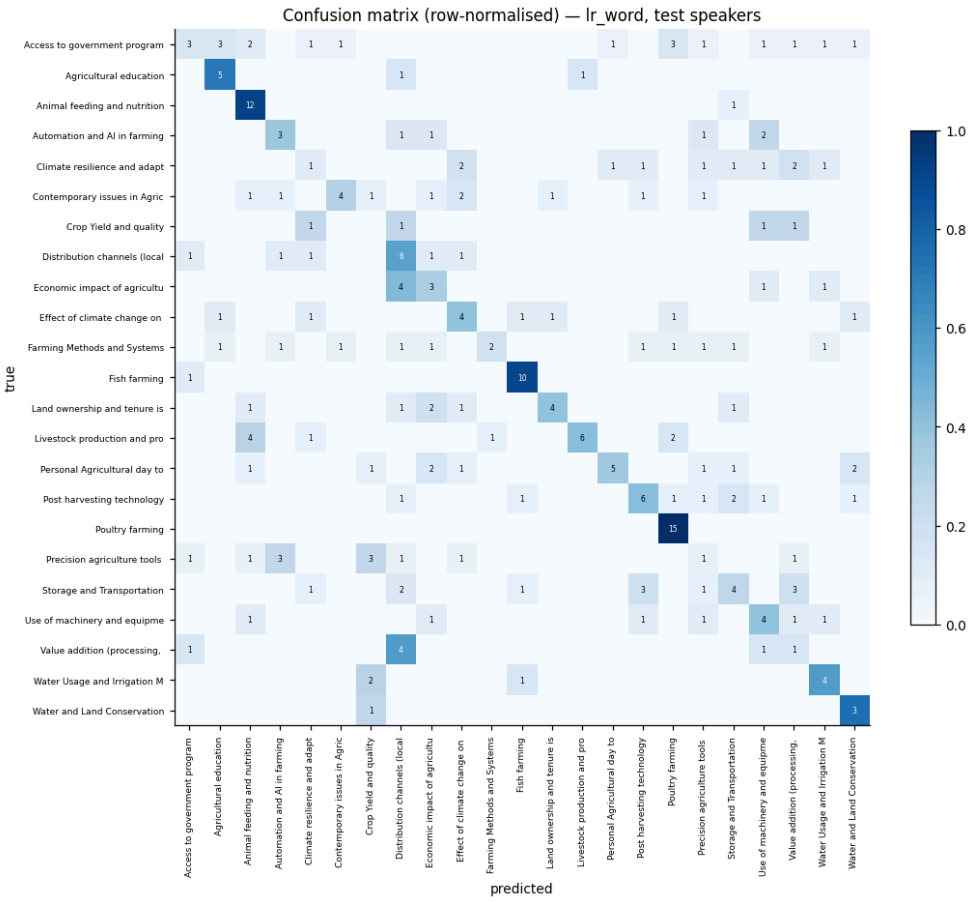
Validation macro-F1 per epoch for every transformer run.

7.2 The selected model on the test set

On the five unseen test speakers the selected model reaches macro-F1 0.396, macro precision 0.430, macro recall 0.424, accuracy 0.424 and weighted F1 0.409 (validation: macro-F1 0.427, accuracy 0.468). The small validation-test gap suggests the selection did not overfit the validation speakers.

Best classes	F1	Worst classes	F1
Fish farming	0.800	Crop yield and quality (4 test examples)	0.000
Poultry farming	0.789	Precision agriculture tools	0.091
Animal feeding and nutrition	0.667	Climate resilience and adaptation	0.111
Agricultural education	0.588	Value addition (processing, packaging)	0.118
Livestock production	0.571	Access to government programmes	0.231

Classes with distinctive vocabulary (*rech*, fish; *gweno*, chicken) are easy; broad or overlapping classes are hard. The full per-class table is in `reports/REPORT_TABLES.md` and notebook section 13.



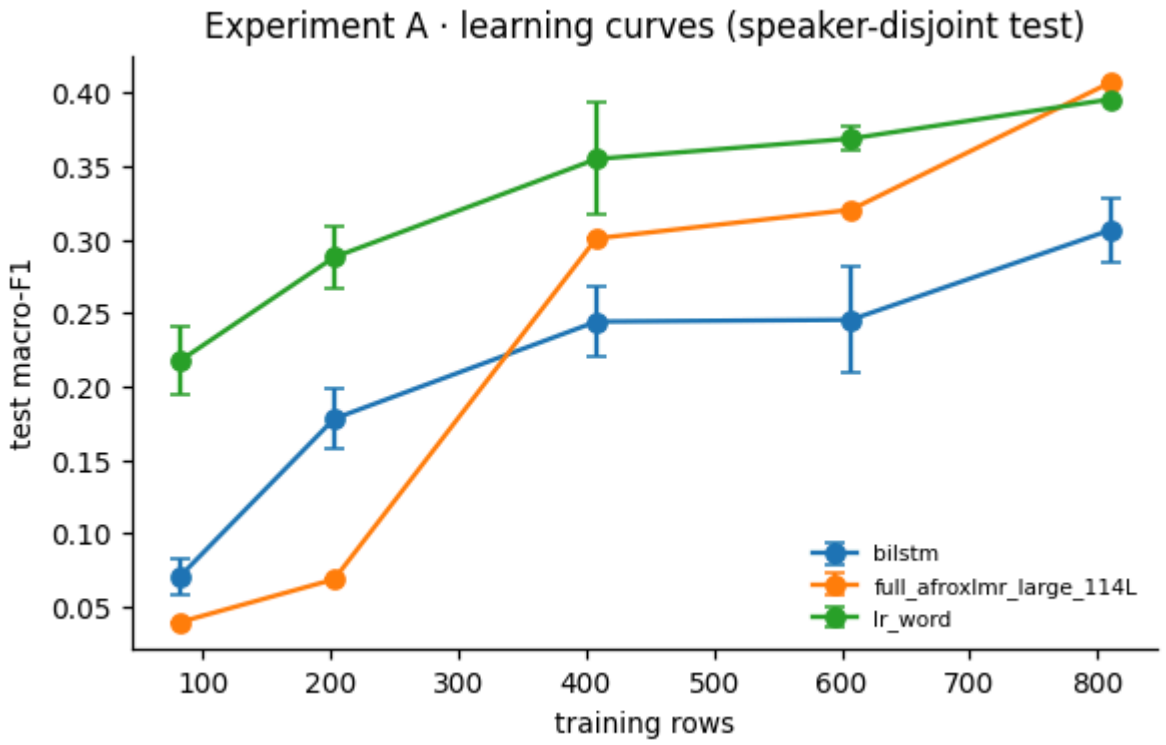
Row-normalised confusion matrix of the selected model on the test speakers.

Calibration. The model's probabilities are under-confident: mean top-class probability 0.110 against accuracy 0.424, expected calibration error 0.314 (10 bins). This does not affect the predicted class, but raw probabilities would mislead users; see section 9.

7.3 Data size, normalisation and class weighting (Experiments A, B, ablation)

Training rows	82	203	408	607	810
TF-IDF + LR	0.217	0.288	0.355	0.369	0.396
BiLSTM	0.070	0.178	0.244	0.245	0.307
AfroXLMR full (1 seed)	0.039	0.069	0.301	0.320	0.407

Test macro-F1. All three models are still improving at full size. Logistic regression is the most data-efficient; the transformer is ineffective below about 200 examples but has the steepest final step (+0.087), suggesting it would pull ahead with more data. The transformer curve uses one seed and should be read as indicative.



Learning curves on the speaker-disjoint test set.

Normalisation has no material effect: the three text variants give test macro-F1 0.391 to 0.396 for logistic regression and 0.307 to 0.320 for the BiLSTM, spreads smaller than the BiLSTM's seed-to-seed variation (about 0.02). `normalized` was selected on validation (0.427 vs 0.424 and 0.423).

Class weighting helps modestly and consistently: logistic regression 0.381 to 0.396 macro-F1 and 0.400 to 0.424 macro recall; BiLSTM 0.297 to 0.307 (within seed noise). With a class ratio of only 3.3 a large effect was not expected.

7.4 Speaker leakage, dialect and code-switching (Experiments F, D, E)

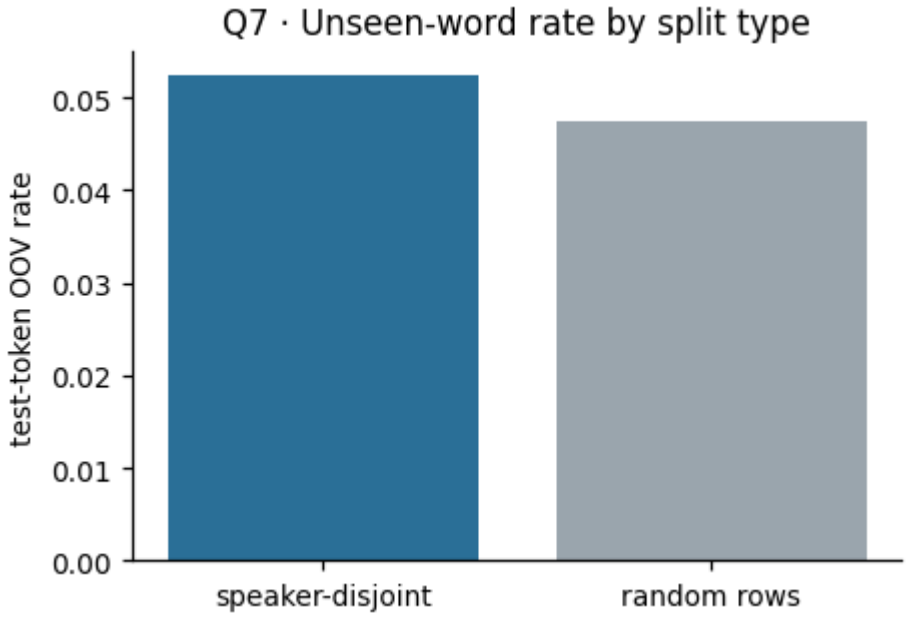
Cross-validation on development speakers	Random rows	Speaker-grouped	Inflation
TF-IDF + LR	0.404	0.368	+0.036
BiLSTM	0.280	0.259	+0.020

A random split overstates performance, modestly. Consistent with this, unseen-word rates on test tokens are 5.2% under the speaker split and 4.8% under a random split (figure 4): speakers share a common agricultural vocabulary.

Dialect. On speaker-grouped out-of-fold predictions for 18 development speakers, logistic regression scores 0.373 on Milambo and 0.372 on Nyanduat speakers (BiLSTM 0.287 and 0.285); on the official test set all models' dialect intervals overlap. Per-speaker accuracy ranges from 0.28 to 0.49, far wider than any dialect difference. With 9 Milambo speakers in total, a small dialect effect cannot be ruled out.

Code-switching. Logistic regression scores 0.367 on Dholuo-only and 0.355 on code-switched transcripts; the BiLSTM 0.279 and 0.302; all intervals overlap. Code-switching

rates are similar by dialect (17.5% Milambo, 15.6% Nyanduat) but vary between speakers. Code-switched words such as *robot*, *drone* and *store* are themselves strong topic cues (notebook section 7, Q8).



Unseen-word rate on test tokens under speaker-disjoint and random splits.

7.5 Prompt overlap (Experiment H)

Model	Seen prompt (n = 101)	Unseen prompt (n = 149)
TF-IDF + LR (word), selected	0.421	0.361
TF-IDF + SVM	0.501	0.344
TF-IDF + LR (char)	0.491	0.292
LoRA AfroXLMR	0.413	0.328
AfroXLMR full	0.443	0.329
BiLSTM	0.378	0.234

Retraining without the 101 training rows whose prompts occur in test	Speaker-disjoint	Speaker + prompt disjoint
TF-IDF + LR (810 to 709 rows)	0.396	0.311
BiLSTM	0.307	0.274

Every model performs better on answers to prompts seen in training, and removing shared prompts costs logistic regression 0.085 macro-F1. Character n-grams, which can match long stretches of a question verbatim, show the largest gap. This is the largest effect in the project: part of every headline score comes from recognising question wording, and **the realistic estimate for new questions is about 0.31 to 0.36 macro-F1**. The model card and application report the unseen-prompt figure.

7.6 Existing tools (Experiment I)

System	Uses Dholuo training data	Val macro-F1	Test macro-F1 (95% CI)	Test acc.
NLLB-200 to English + BART-MNLI zero-shot	no	0.244	0.181 (0.133 to 0.219)	0.224
NLLB-200 to English + TF-IDF LR trained on translations	yes	0.423	0.318 (0.260 to 0.363)	0.364
TF-IDF + LR on Dholuo (selected)	yes	0.427	0.396 (0.329 to 0.442)	0.424

All 1,248 transcripts were translated (in chunks of up to 40 words, beam size 2) in 129 seconds. The Dholuo model beats both routes significantly (paired differences +0.214, interval +0.142 to +0.278; and +0.077, interval +0.015 to +0.136). Translations read fluently but lose meaning: in the qualitative sample *oganda* (beans) is rendered as “nation”, and an answer about storing dry maize (*oduma*) becomes storing “spices” and “seedlings”. Existing translation tools therefore do not solve this task.

7.7 Fine-tuning strategy (Experiment G)

Strategy (AfroXLMR)	Trainable parameters	Test macro-F1	Time per run	Peak GPU memory
Frozen encoder + LR	23,575	0.227	104 s	n/a
LoRA r = 16	2,646,039	0.436 ± 0.060	331 s	4.5 GB
Full fine-tuning	303,911,959	0.416 ± 0.025	610 s	7.2 GB

LoRA gives the best mean with about half the time and memory of full fine-tuning; a likely reason is that training 2.6M rather than 304M parameters leaves less room to overfit 810 examples. Its larger seed-to-seed spread is the trade-off. The frozen general-purpose representation is not sufficient on its own.

8. Error analysis and discussion

8.1 Quantitative error analysis

The selected model misclassifies 144 of 250 test answers (57.6%). Each test answer was tagged with objective flags defined before inspecting errors:

Flag	Rows	Error rate with flag	Error rate without	Relative risk
Prompt unseen in training	149	0.624 (0.54 to 0.70)	0.505 (0.41 to 0.60)	1.24
Long answer (at least training Q3 length)	142	0.620	0.519	1.20
Gold class has low training support	61	0.656	0.550	1.19
Code-switched	52	0.635	0.561	1.13
High unseen-word rate	123	0.585	0.567	1.03
Milambo speaker	177	0.565	0.603	0.94
Female speaker	116	0.543	0.604	0.90
Editor-corrected transcript	115	0.539	0.607	0.89
Short answer (at most training Q1 length)	21	0.429	0.590	0.73

Unseen prompts, long answers and rare classes raise the error rate; dialect, gender and editorial correction do not. Confusions follow meaning: the most frequent confusion pairs are distribution channels with economic impact and with value addition, post-harvest technology with storage and transportation, automation and AI with precision agriculture tools, and animal feeding with livestock production. Across all 253 class pairs, lexical similarity between classes (cosine similarity of training TF-IDF centroids) correlates with confusion rate (Spearman rho = 0.30, p = 1.1e-6). 77 of 250 test answers (30.8%) are misclassified by every model in the study, which points to properties of the data (overlapping topic definitions, answers that do not match their question’s label) rather than to any one model. Per-speaker error rates are similar (0.50 to 0.62).

8.2 Qualitative analysis

Twelve test examples were drawn at random with a fixed seed from six groups (confident and correct, unsure and correct, confident and wrong, unsure and wrong, code-switched, short). The readings below are based on the question asked, the transcript and the NLLB English translation, and should be confirmed by a Dholuo speaker before being quoted as linguistic fact.

- **Correct predictions are driven by topic words.** Both poultry answers rest on *gweno* / *gwen* (chicken), both fish-farming answers on *rech* (fish), and the irrigation answer on *olo pii* (draw water), as shown by word-occlusion analysis. A robotics answer is classified through the English loanword *robot*, an instance of code-switching helping.
- **Neighbouring topics overlap.** An answer on how small-scale farmers benefit from processing *or storing* produce (label: value addition) is predicted as distribution channels; an answer on drying maize so it keeps (label: storage and transportation) is

predicted as value addition. Drying, storing, processing and selling are stages of one chain, and the answers mention several.
- **The label follows the question, the content something else.** A question on why farmers seek help with pests and diseases of crops and livestock carries the label “access to government programmes”, but the answer discusses animals and crops, and the model predicts animal feeding. A question on how growing trade (*ohala*) helps farmers is labelled value addition, while the answer discusses money and markets; the model’s top three (government programmes, economic impact, distribution channels) are arguably closer to what was said.
- **Rare loanwords.** An answer about protecting stored crops with a locked store and dogs (label: storage) is predicted as distribution channels, with storage second; the deciding word appears to be a Kiswahili loanword inside a code-switched span, too rare in training to carry weight.

8.3 Discussion

Three conclusions follow. First, in this data regime a well-tuned linear model is a strong, honest choice: pretrained encoders tie with it on test, need minutes of GPU time per run instead of seconds on a CPU, and are less stable across seeds. Second, African-language pretraining is what makes the transformers viable at all; the same architecture without it fails. Third, and most important for anyone building on this dataset, evaluation must control for prompt overlap as well as speakers, because question echo inflates scores by roughly 0.03 to 0.09 macro-F1 (section 7.5). The error analysis indicates that further gains depend as much on the labels (answer-level annotation, merged or redefined adjacent sub-domains) as on the model.

9. Streamlit application design and implementation

Architecture. The application separates the interface from inference. `src/preprocessing.py` reproduces the training normalisation exactly (verified by a test that executes the notebook’s preprocessing source on the same inputs). `src/prediction.py` loads `label_map.json` and `model.joblib` from `models/final/` (or `DHOLUO_MODEL_DIR`), checks that the model kind, pipeline structure and label count match, normalises input with the text variant recorded at training time, and returns a ranking over all 23 sub-domains. `app.py` is a thin Streamlit layer.

Behaviour.

- Accepts a Dholuo transcript; transcription markers such as `[cs]...[cs]`, `[pause]` and `[?]` are handled as in training.
- Shows the most likely sub-domain, described as the dataset’s question-associated topic rather than a judgement of what the answer is about, and the next two.
- Shows confidence percentages only when `label_map.json` records a temperature fitted on validation data. Because the exported model’s raw probabilities are under-confident (ECE 0.314), the application currently shows the ranking and explains why percentages are withheld; raw scores are available in a collapsed panel labelled as uncalibrated. `scripts/calibrate_temperature.py` fits the temperature on the validation speakers from the exported split file (temperature scaling; Guo et al., 2017).

- Rejects empty, word-less or oversized input with a clear message; warns about inputs shorter than 10 words (training minimum 16), longer than 300 words (training maximum 158), or with fewer than 60% of words known to the model, and states that every input receives one of the 23 topics.
- Reports missing or incompatible model files with instructions instead of failing, and displays the verified evaluation figures, limitations and responsible-use guidance in a sidebar.

Validation performed. 31 automated tests pass (preprocessing parity, prediction, input validation, artifact errors, calibration script, and the app through Streamlit's testing harness). The application was also launched headless and driven in a browser (Playwright/Chromium): classification, warnings and empty-input handling behaved as intended. These checks used a small stand-in model with the exported model's structure, because the trained model is produced on Kaggle and is not stored in the repository; the application has not yet been run against the real exported model in this environment (Appendix C).

10. Limitations and reliability considerations

- **Question-derived labels.** The model predicts the question-associated sub-domain; some answers drift or span topics (section 8.2).
- **Prompt echo.** Headline scores include a benefit from seen questions; about 0.31 to 0.36 macro-F1 is the realistic estimate for new questions.
- **Small, narrow speaker base.** 23 speakers, 13 in training, 18 of them aged 18 to 29 and one over 40; one transcription team; agriculture only. The test set has 5 speakers, so subgroup estimates are imprecise.
- **Dialect evidence is weak.** 9 Milambo speakers (3 in training) and inconsistent dialect metadata.
- **Moderate accuracy.** About 4 in 10 test answers are correct; 31% of test answers defeat every model.
- **Calibration.** Raw probabilities are under-confident until temperature scaling is applied.
- **Single-seed transformer learning curve** and a single split; results could shift with a different selection of test speakers.
- **Closed label set.** No out-of-domain detection; non-Dholuo or off-topic text still receives a topic.
- **Text only.** The application does not accept audio.

11. Ethical considerations and responsible use

Voice recordings are biometric data. The project uses only transcripts and pseudonymous speaker identifiers, never attempts to identify contributors, writes no transcripts to shared outputs, and does not redistribute the gated dataset; the repository contains code, configuration, split identifiers and instructions. The NOODL-1.0 licence permits open use in developing countries and asks users elsewhere to share benefits with the contributing communities. Because the TF-IDF vocabulary is drawn from the transcripts, the trained model should not be published without checking the licence and access conditions.

The system is suitable only as a low-stakes routing or search aid with a person reviewing the output. It must not be used for decisions about people, such as eligibility for subsidies, credit or programmes. Its training population (mostly young adults) means it may serve older farmers, other regions and other dialect groups less well, and it should not be described as understanding Dholuo in general.

12. Conclusion and future improvements

Completed work. A data audit and task justification; a speaker-disjoint benchmark of 23 agricultural sub-domains; a comparison of eight model configurations and two floors under one protocol with seeds and bootstrap statistics; ten controlled experiments; quantitative and qualitative error analysis; and a tested Streamlit application around the selected model.

Answer to the research question. On unseen speakers, the best models reach about 0.40 to 0.44 macro-F1 on 23 classes (stratified random baseline 0.035; majority class 0.005), with a well-tuned TF-IDF logistic regression statistically tied with LoRA-tuned AfroXLMR. Performance grows with data for every model, does not depend measurably on normalisation, dialect or code-switching at this scale, requires African-language pretraining for transformers to work, exceeds what existing translation tools achieve, and falls to about 0.31 to 0.36 for questions not seen in training.

Future improvements (not yet done).

1. Fit and record the validation temperature for the deployed model (script provided) and re-check calibration on test.
2. Collect more speakers, especially Milambo speakers and older farmers; all learning curves were still rising.
3. Annotate the topic of each answer, and merge or redefine adjacent sub-domains (value chain stages; automation and precision tools).
4. Use prompt-disjoint as well as speaker-disjoint splits in future data releases.
5. Fine-tune ASR on the same speaker split so the application can accept speech.
6. Continue pretraining AfroXLMR on the unlabelled image-prompt transcripts, 61% of whose word types do not occur in the training data.
7. Evaluate on non-agricultural Dholuo and add out-of-domain detection.

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Appendix A. Traceability of results

Report section	Source in `notebooks/dholuo_topic_classification_final.ipynb`
5.2 to 5.6	Section 3 (audit, contributors, label validity, prompt echo), section 5 (cleaning log)
6.2	Section 6 (split search, split summary, leakage checks)
6.3	Sections 10, 11, 11.3 (model summary, configurations, tokenizer fertility), 18b
7.1	Section 12 (Experiment C table and paired bootstrap)
7.2	Section 13 (per-class report, confusion matrix, calibration line)
7.3	Sections 9, 14.1, 14.2, 14.3
7.4	Sections 14.4, 14.5, and section 7 (Q6, Q7)
7.5	Section 14.6
7.6	Section 14.7
7.7	Section 11.2
8.1	Section 16
8.2	Section 15 and the “Reading the examples” cell
9	`src/`, `app.py`, `tests/` in this repository

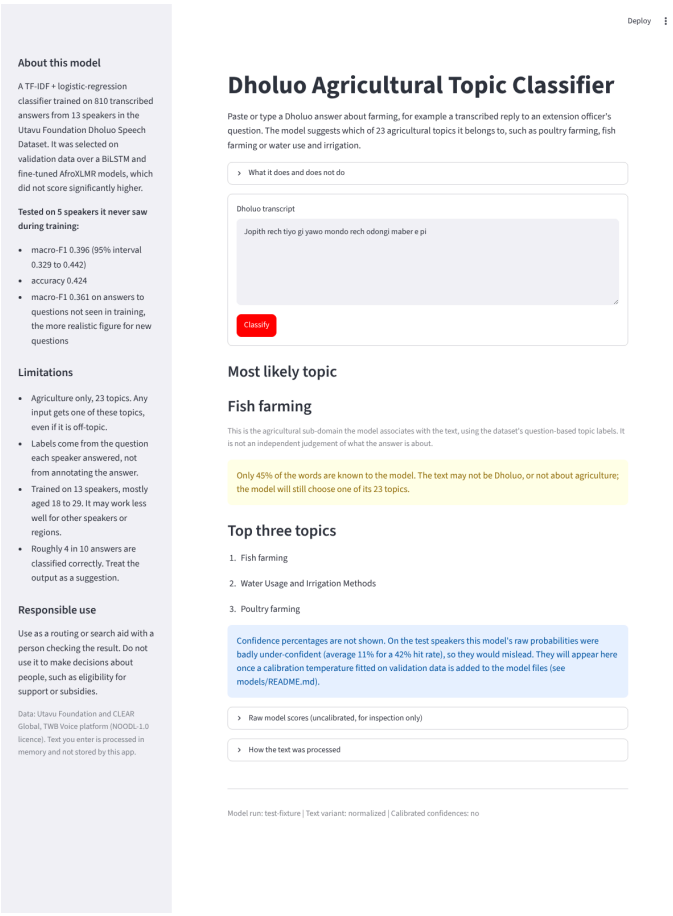
The exported tables are reproduced in `reports/REPORT_TABLES.md`; figures in `reports/figures/` were extracted from the notebook’s saved outputs.

Appendix B. Hyperparameters

Setting	Value
Split seed; training seeds	42; 13, 42, 2026
Split search	20,000 candidate speaker groupings; target 20% test, 15% validation
Class eligibility	at least 25 recordings and 5 speakers
TF-IDF LR (word), selected	unigrams, min_df 1, sublinear tf, case preserved, C = 1, class_weight balanced, L-BFGS (max_iter 5,000)
TF-IDF SVM	word 1-2-grams + char_wb 2-5-grams, C = 0.1, balanced; temperature 0.080 (validation)
BiLSTM	embedding 128, hidden 128 per direction, dropout 0.4, AdamW lr 2e-3, weight decay 1e-4, batch 32, max 40 epochs, patience 6, max length 200 tokens, min frequency 2, gradient clip 1.0
Transformers (full)	lr 2e-5, batch 8 x accumulation 2, warm-up 10% then linear decay, weight decay 0.01, max 20 epochs, patience 5, max length 256, fp16, frozen input embeddings, gradient checkpointing
LoRA	r 16, alpha 32, dropout 0.1, targets query and value, lr 2e-4
Experiment F	5-fold cross-validation on development speakers, repeated for each seed
Experiment I	NLLB-200 distilled 600M, `luo_Latn` to `eng_Latn`, chunks of at most 40 words, beam 2, max 128 new tokens; BART-large-MNLI, template “This text is about {}.”
Bootstrap	2,000 resamples

Appendix C. Validation status of the software

Check	Status
Unit and integration tests (`python -m pytest`)	31 passed (Python 3.13, scikit-learn 1.6.1, Streamlit 1.66.0)
Preprocessing parity with the notebook	Passed on the notebook’s printed examples and 13 edge cases
Streamlit app launched headless and exercised in Chromium	Passed with the stand-in model
App with the real exported model	Not performed here: the model is produced on Kaggle and not stored in the repository
Calibration script on real data	Not performed here: the gated dataset cannot be reached from the build environment; tested offline with synthetic data



The application interface, rendered with the test stand-in model. Predictions and the run name shown are from that stand-in, not from the trained model.